

Katherine Papciak

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EDUCATION

New York University, Vilcek Institute of Graduate Biomedical Sciences and Grossman School of Medicine, New York, NY
(Expected) August 25, 2025 | **MSc. in Biomedical Informatics (Artificial Intelligence Concentration)**

Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Generative AI, and Data Science, Bioinformatics, Methods in Quantitative Biology, Healthcare and AI

Virginia Polytechnic Institute and State University, College of Science, Blacksburg, VA

May 18, 2019 | **BSc. in Biological Sciences (Biomedical Concentration), Statistics (Minor)**

Relevant Coursework: Calculus, Methods of Regression Analysis, Nonparametric Statistics, Applied Time Series, Experimental Design, Proteomics Mass Spectrometry, Immunology, Biochemistry, Organic Chemistry, Human Genetics, Virology, Physics

TECHNICAL SKILLS

Programming Languages: R, Python, SQL, SAS, GraphPad/Prism, Minitab

Operating Systems: MacOS, Microsoft Windows, Linux/UNIX, Ubuntu, HPC/SLURM environments

Computational Tools/Libraries: pandas, NumPy, tidyverse, ggplot2, Seaborn, matplotlib, TensorFlow, PyTorch

Certifications: Lean Six Sigma Yellow Belt

EXPERIENCE

Senior Scientific Administrative Coordinator, Farese and Walther Lab, Sloan Kettering Institute, New York, NY

January 23, 2023 to August 23, 2024 | Dr. Tobias Walther, PhD and Dr. Robert Farese Jr., MD

- Developed analytical methods to investigate and visualize triglyceride accumulation in motor neurons using advanced microscopy and lipidomics techniques, applying statistical analysis to large datasets.
- Implemented data storage and management infrastructure for research data on HPC cluster, ensuring data integrity and accessibility across multiple projects.
- Created dashboards and visualizations to track project goals, budgetary issues, scientific output, and resource allocation.
- Assisted PIs with preparation of research proposals and manuscripts, translating complex scientific findings into accessible formats.

Genetic Counselor Assistant, Sema4/GeneDx, Stamford, CT

August 23, 2021 to January 12, 2023 | Caroline Sotile MS, CGC

- Created structured data pipelines to organize and standardize patient information, ensuring data integrity and HIPAA compliance.
- Generated and improved standard operating procedures for hereditary cancer, carrier screening, and prenatal counseling workflows.
- Collaborated with genetic counselors to develop educational materials based on patient data insights.

Research Technician, Beyaz Lab and Meyer Lab, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY

June 10, 2019 to August 18, 2021 | Dr. Semir Beyaz, PhD and Dr. Hannah Meyer, PhD

- Managed and analyzed data from longitudinal diet studies in mice, including metabolic, genomic, and flow cytometry datasets. Applied statistical methods to analyze experimental results, including GTT/ITT, fecal metagenomics, and tissue metabolomics.
- Coordinated collaborative projects across multiple institutions, effectively communicating technical information to diverse stakeholders.
- Developed protocols for laboratory experiments including high-throughput FACS and scRNA-seq, optimizing data collection processes.

Undergraduate Research Assistant, Virginia Tech, Blacksburg, VA

January 24, 2019 to May 10, 2019 | Dr. Christopher Lawrence, PhD

- Applied bioinformatics techniques to analyze gene expression data through Gene Set Enrichment Analysis (GSEA) and Ingenuity Pathway Analysis (IPA), developing cytokine signaling pathway visualization for improved data interpretation.

Flex/Training Lab Intern, Sartorius Stedim Biotech, Bohemia, NY

June 4, 2018 to August 24, 2018 | Mr. Mark Walker

- Led training sessions on quality control data analytics and biopharmaceutical equipment, translating complex technical information for diverse stakeholders and ensuring compliance with industry standards.

Informatics Intern, BioTherapeutics Inc., Blacksburg, VA

August 28, 2017 to May 14, 2018 | Dr. Pinyi Lu, PhD

- Designed data mining algorithms in R to leverage clinical combination therapy data for inflammatory bowel disease and Crohn's disease patients. Conducted preliminary statistical analyses to assess treatment efficacy, identifying potential patterns in patient outcomes.

RECENT PROJECTS

Evaluation of Large Language Model (LLM) Approaches for Cancer Progression Identification from Clinical Notes - Master's Thesis

- Led comprehensive research evaluating advanced LLM methodologies to automate lung cancer progression detection from clinical text data - Developed systematic approach to evaluate multiple LLMs (LLaMA 3, DeepSeek, and OpenBioLLM) across zero-shot, few-shot, and fine-tuned implementations.
- Created specialized prompt engineering techniques including chain-of-thought reasoning and role-based prompting optimized for clinical context.
- Implemented Retrieval-Augmented Generation (RAG) with medical literature knowledge base and dense vector embeddings.
- Designed a comprehensive evaluation framework using accuracy, precision, recall, F1, AUC, and clinically-relevant metrics.
- Conducted statistical discordance analysis between model outputs and human annotations using advanced methods (Kendall's concordance, IRT, Bayesian modeling).

Privacy-Preserving Synthetic Whole Genome Data Using Deep Generative Models - Deep Learning Final Project

- Led development of advanced deep generative models to synthesize privacy-protected whole genome data while preserving statistical properties for biomedical research.
- Implemented and compared three architectures: Developed modified Wasserstein GAN with gradient penalty using differential privacy techniques, created specialized Variational Autoencoder with linkage disequilibrium (LD) aware encoding and haplotype-preserving features, implemented novel Diffusion Model adapted for discrete genomic data with ancestry-conditional generation.
- Designed comprehensive evaluation framework measuring: Statistical fidelity (Minor allele frequency distribution, LD pattern preservation), privacy protection (membership inference attack resistance), utility preservation (GWAS reproducibility, predictive modeling performance).